FOURIER SERIES | Applications to BVPs & PDEs

Engineering Mathematics · Separation of Variables & Eigenfunction Expansions

1. Conceptual Foundation

A **Fourier Series** represents a periodic function (or a function on a finite interval) as an infinite sum of sines and cosines (or complex exponentials) of harmonically related frequencies.
Core idea: On an interval of length L, the set $\{1, \cos(2\pi nt/L), \sin(2\pi nt/L)\}$ forms an orthogonal basis. Any reasonably well-behaved function can be expanded in this basis.
"Fourier Series converts a spatial (or temporal) profile into discrete harmonic coefficients that are easy to manipulate under linear differential operators."

Link to PDEs: Separation of variables reduces many linear PDEs with homogeneous boundary conditions to Sturm–Liouville eigenvalue problems whose eigenfunctions are precisely the Fourier basis. The series coefficients of the initial/boundary data become the coefficients of the PDE solution.

2. Full Fourier Series (Period $2L$)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Coefficients (Euler–Fourier formulas):

$$a_0 = \frac{1}{L} \int_{-L}^{L} f(x)\, dx,$$
$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos \frac{n\pi x}{L}\, dx,$$
$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin \frac{n\pi x}{L}\, dx.$$

Complex form:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}, \qquad c_n = \frac{1}{2L} \int_{-L}^{L} f(x) e^{-in\pi x/L}\, dx.$$

3. Even/Odd Extensions & Half-Range Series

On $(0, L)$:
- **Even extension** $\Rightarrow$ cosine series only ($b_n = 0$).
- **Odd extension** $\Rightarrow$ sine series only ($a_n = 0$).

Half-range sine series (odd extension):

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L}\, dx.$$

Half-range cosine series (even extension):

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L}\, dx.$$

Choice rule for PDEs: Match the extension to the homogeneous boundary conditions (Dirichlet $\rightarrow$ sine; Neumann $\rightarrow$ cosine).

4. Convergence & Gibbs Phenomenon

If f is piecewise smooth on $[-L, L]$:
- At a continuity point the series converges to $f(x)$.
- At a jump discontinuity it converges to the average of left and right limits.
- Near a jump the partial sums overshoot by $\approx 9\%$ of the jump height (**Gibbs phenomenon**); the overshoot does not disappear as $N \to \infty$, it merely becomes narrower.

Term-by-term differentiation is legitimate only under stronger conditions (e.g., continuous f' or uniform convergence of the differentiated series).

5. Useful Properties

Property	Statement
Linearity	$\mathrm{FS}\{\alpha f + \beta g\} = \alpha \mathrm{FS}\{f\} + \beta \mathrm{FS}\{g\}$
Differentiation	If f continuous & f' piecewise smooth, differentiate term-by-term
Integration	Integrate term-by-term; constant of integration from a_0
Parseval	$\frac{1}{L} \int_{-L}^{L} \lvert f \rvert^2\, dx = \frac{\lvert a_0 \rvert^2}{2} + \sum (\lvert a_n \rvert^2 + \lvert b_n \rvert^2)$

6. Separation of Variables — Algorithm

For a linear homogeneous PDE with homogeneous BCs on a finite interval:

1. Assume product solution $u(x,t) = X(x)T(t)$ (or $X(x)Y(y)$).
2. Substitute into the PDE; separate variables $\Rightarrow$ two ODEs linked by a separation constant λ.
3. Impose homogeneous BCs on the spatial ODE $\Rightarrow$ Sturm–Liouville eigenvalue problem.
4. Find eigenvalues λ_n and eigenfunctions $X_n(x)$ (usually sines or cosines).
5. Solve the temporal (or second spatial) ODE for each n.
6. Form the general series solution $u = \sum c_n X_n(x)\, T_n(t)$.
7. Determine coefficients c_n from the remaining initial or boundary condition by Fourier expansion.

7. Heat Equation (1-D)

$$u_t = k u_{xx}, \qquad 0 < x < L,\ t > 0$$

Dirichlet BCs $u(0,t) = u(L,t) = 0$:

$$u(x,t) = \sum_{n=1}^{\infty} b_n e^{-(n\pi/L)^2 kt} \sin\frac{n\pi x}{L},$$

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx \quad \big(u(x,0) = f(x)\big).$$

Neumann BCs $u_x(0,t) = u_x(L,t) = 0$: cosine series; the $n = 0$ term is the conserved average temperature.

Key observation: Higher modes decay exponentially faster $\rightarrow$ solution smooths and approaches the steady state.

8. Wave Equation (1-D)

$$u_{tt} = c^2 u_{xx}, \qquad 0 < x < L,\ t > 0$$

Dirichlet ends $u(0,t) = u(L,t) = 0$:

$$u(x,t) = \sum_{n=1}^{\infty}\Big(A_n\cos\frac{n\pi ct}{L} + B_n\sin\frac{n\pi ct}{L}\Big)\sin\frac{n\pi x}{L}.$$

Initial displacement $u(x,0) = f(x) \Rightarrow A_n =$ Fourier sine coefficients of f.

Initial velocity $u_t(x,0) = g(x) \Rightarrow B_n = \dfrac{L}{n\pi c}\times$ (sine coefficients of g).

Interpretation: Standing waves; frequencies are integer multiples of the fundamental $c/(2L)$.

9. Laplace Equation (Rectangle)

$$u_{xx} + u_{yy} = 0, \qquad 0 < x < a,\ 0 < y < b$$

Typical BCs: three sides zero, one side prescribed $u(x,0) = f(x)$ (or similar).
Expand in the direction of the homogeneous BCs (usually a sine series in x). The y-dependence becomes $\sinh(n\pi y/a)$ (or linear combination of $\sinh$ and $\cosh$) chosen to satisfy the remaining BC.

Selection rule:

- Homogeneous in $x \rightarrow$ expand in x, hyperbolic functions in y.
- Homogeneous in $y \rightarrow$ expand in y, hyperbolic functions in x.

10. Choosing Sine vs Cosine from BCs

Homogeneous BCs	Natural series
$u(0) = u(L) = 0$ (Dirichlet)	Sine series
$u_x(0) = u_x(L) = 0$ (Neumann)	Cosine series
$u(0) = u_x(L) = 0$ (mixed)	Quarter-range / shifted sine
Periodic BCs	Full Fourier series

Always expand the non-homogeneous data in the eigenfunctions that already satisfy the homogeneous boundary conditions.

11. Common Errors — Quick Check

- Using a full-range series when the problem is posed only on $(0, L)$.
- Choosing sine series for Neumann conditions (or vice versa).
- Forgetting the steady-state / particular solution when BCs are non-homogeneous.
- Differentiating a Fourier series term-by-term without checking convergence conditions.
- Incorrect limits or missing factor $2/L$ in half-range coefficients.
- Ignoring the $n = 0$ (constant) term in cosine series for the heat equation.
- Applying separation of variables to a nonlinear PDE.
- Confusing the roles of the two independent variables when selecting the expansion direction for Laplace's equation.